2025-26 S1 – BIO220 PROGRAM

Week	Lecture	Tutorials and Labs	Coursework
Week 1	Introduction to computational Biology.		
Week 2	Charges and Coulomb law. Electrostatic potential.		
Week 3	Atoms and molecules. The Schrodinger equation		
Week 4	Molecular mechanics. Force Fields.		Coursework 1 (in class test): Molecular Physics.
Week 5	Structure of proteins and nucleic acids.	Tutorial and Lab: Molecular visualization with VMD.	
Week 6	Structural models and protein folding. Homology modeling and Alphafold.	Tutorial and Lab: Homology modeling.	Coursework 2 (laboratory): Homology modeling and molecular visualization.
Week 7	Molecular docking.		
Week 8	Energy minimization. Normal modes analysis.		Coursework 2 deadline
Week 9	Molecular surfaces and electrostatics.	Tutorial and Lab: Molecular docking, normal modes.	
Week 10	Free energy and solvation.		
Week 11	Small molecules and chemical spaces. Chemoinformatics.	Tutorial and lab: Protein/Ligand modeling.	Coursework 3 (laboratory): Protein/Ligand modeling.
Week 12	Drug libraries, screening and selection.		
Week 13	Recap.		Coursework 3 deadline

BIO220 exam structure

Open questions (4) and closed book.

2 hours time – 30 minutes for each question.

The question are designed to test your understanding of the material covered in the course.
Calculation, if present, are trivial (no problems to be solved, only knowledge questions).
In general, it is not required to memorize formulas, except those that are central for understanding the material.

Important knowledge to have:

1) Structure of biomolecules: proteins, nucleic acids etc.

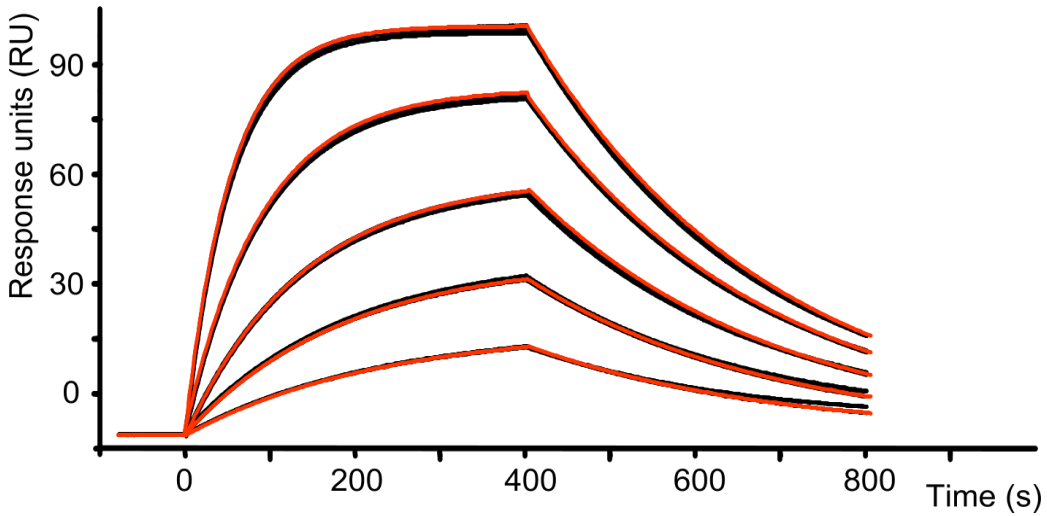
2) Molecular physics: properties of molecules, interactions between atoms, force fields, polarization, solvation ...

3) Practical skills: pdb files, software for visualization, protein structure prediction, docking …

4) Theoretical understanding of folding, docking, sampling …

Some additional words on binding…

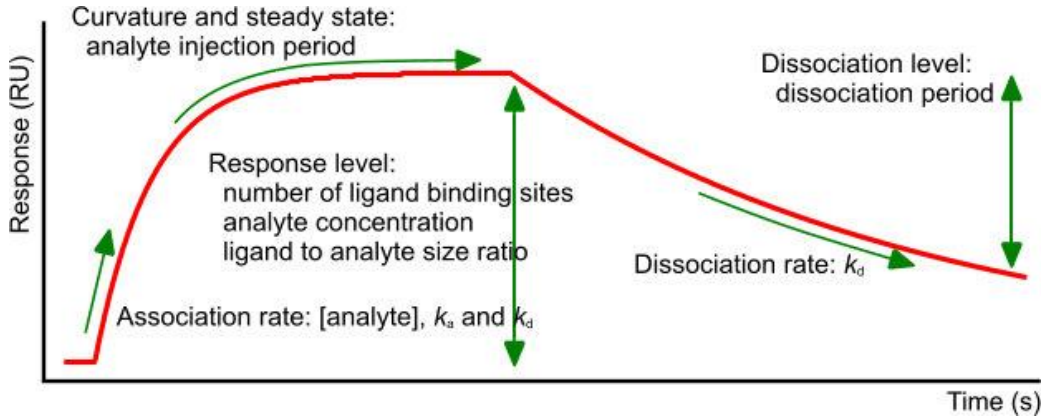
Molecular binding is a dynamical process

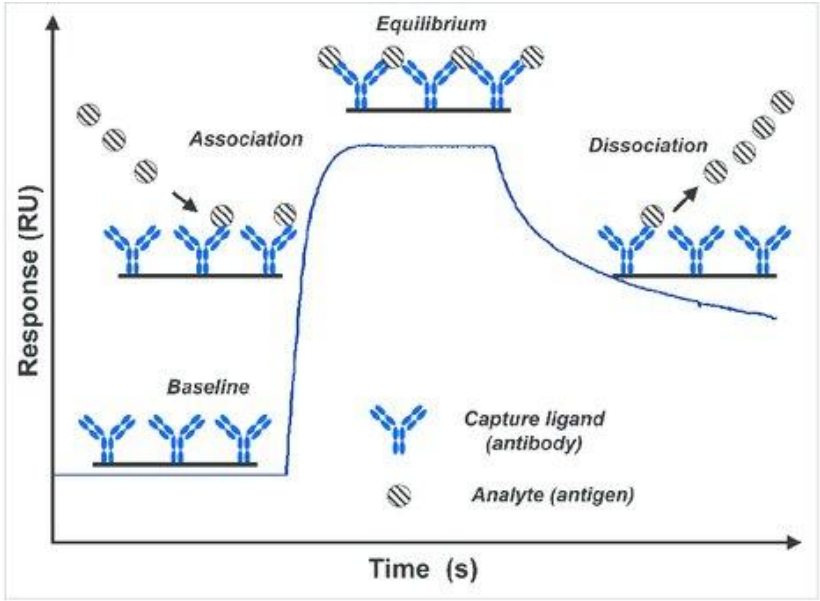


Typical SPR (Surface Plasmon Resonance) experiment.

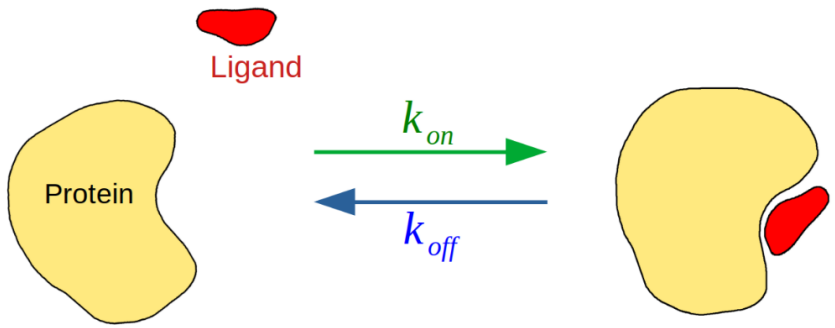
SPR measure binding and unbinding of two molecules as a function of the concentration.

The ratio between the K_{on} and K_{off} is the equilibrium constant (K_D).





Molecular binding is a dynamical process



$$P + L \leftrightharpoons PL$$

$$\frac{d[PL]}{dt} = k_{on}[P][L] - k_{off}[PL]$$

$$\frac{d[P]}{dt} = -k_{on}[P][L] + k_{off}[PL]$$

$$\frac{d[L]}{dt} = -k_{on}[P][L] + k_{off}[PL]$$

$$\frac{d[PL]}{dt} = k_{on}[P][L] - k_{off}[PL]$$

The binding between the Ligand (L) and the protein (P) is proportional to the concentration of the protein and the ligand:

$$\frac{d[PL]}{dt} = k_{on}[P][L]$$

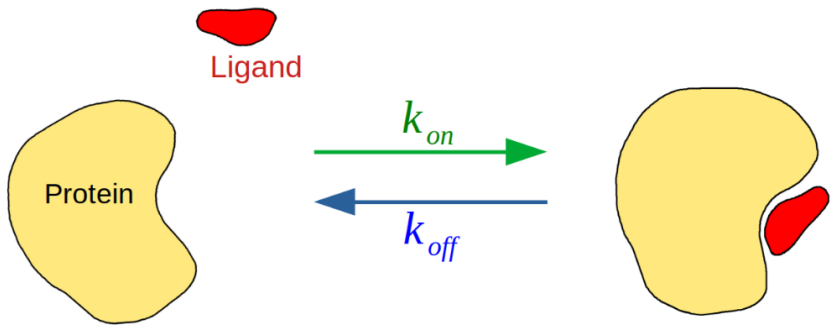
The constant k_{on} measure how easily the binding can happen, and it is measured in 1/M·s.

Unbinding is proportional to the concentration of bounded molecules:

$$\frac{d[PL]}{dt} = -k_{off}[PL]$$

The constant k_{off} measure how easily the unbinding can happen, and it is measured in 1/s.

Molecular binding is a dynamical process



$$P + L \leftrightharpoons PL$$

$$\frac{d[PL]}{dt} = k_{on}[P][L] - k_{off}[PL]$$

$$\frac{d[P]}{dt} = -k_{on}[P][L] + k_{off}[PL]$$

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$$\frac{d[PL]}{dt} = k_{on}[P][L] - k_{off}[PL]$$

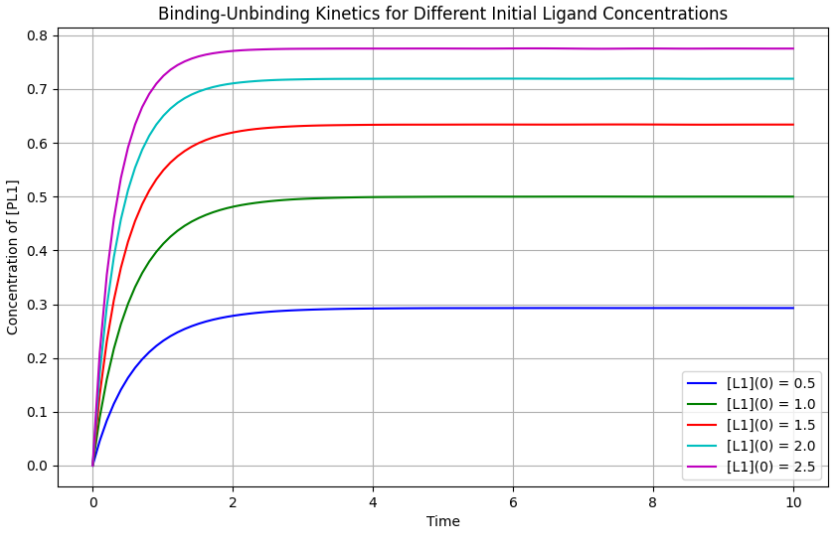
If protein and ligands are in a solutions, a dynamical equilibrium is reached when the association and dissociations processes happen with the same frequency:

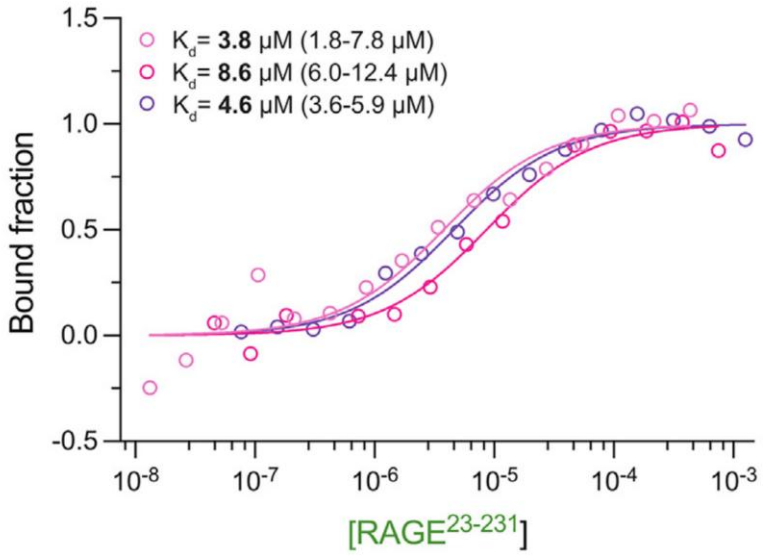
$$\frac{d[PL]}{dt} = 0; \quad k_{on}[P]_e[L]_e = k_{off}[PL]_e$$

The concentrations for which the equilibrium are reached has to follow the relationships:

$$\frac{[P]_e[L]_e}{[PL]_e} = \frac{k_{off}}{k_{on}} = K_D = \frac{1}{K_A}$$

Binding affinity





Binding experiments measure the fraction of target receptors occupied at equilibrium across a range of ligand concentrations.

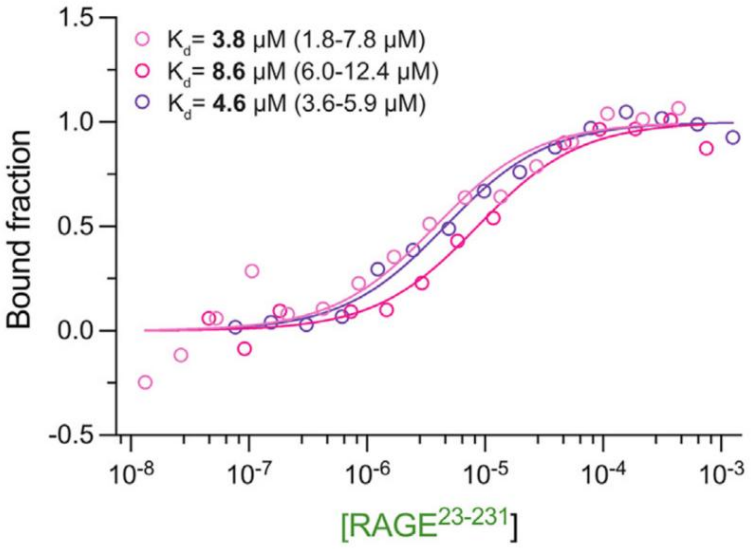
Plotting equilibrium occupancy against ligand concentration generates a characteristic sigmoid saturation curve.

The inflection point of this sigmoid (where 50% of receptors are bound) directly corresponds to the ligand concentration equal to the K_D (see next slide).

Multiple measurements across different concentrations allow fitting to the binding equation, yielding K_D values with statistical confidence intervals.

This equilibrium saturation approach provides a direct experimental readout of binding affinity by identifying the concentration needed for half-maximal target occupancy.

Binding Affinity and Binding Free Energy



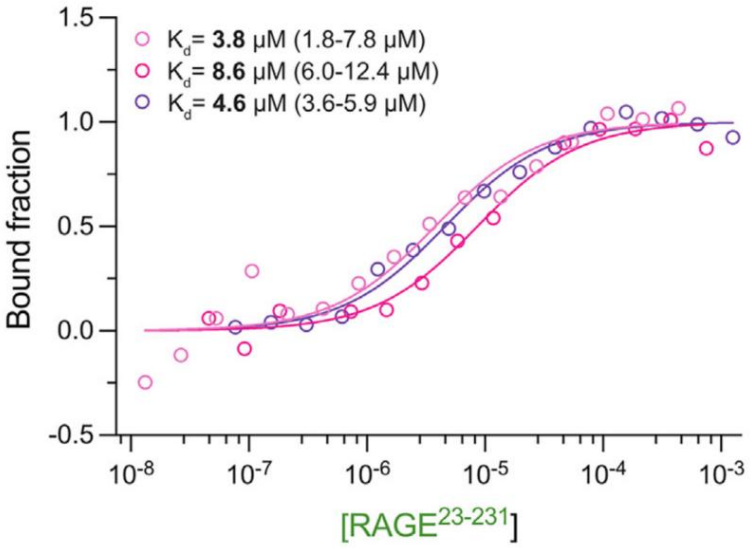
Let us assume we have a generic chemical reaction: : $PL \rightleftharpoons P + L$; we call *dissociation constant* the quantity:

$$K_D = \frac{[P][L]}{[PL]}$$

The binding affinity K_A is the reciprocal of the dissociation constant:

$$K_A = \frac{1}{K_D}$$

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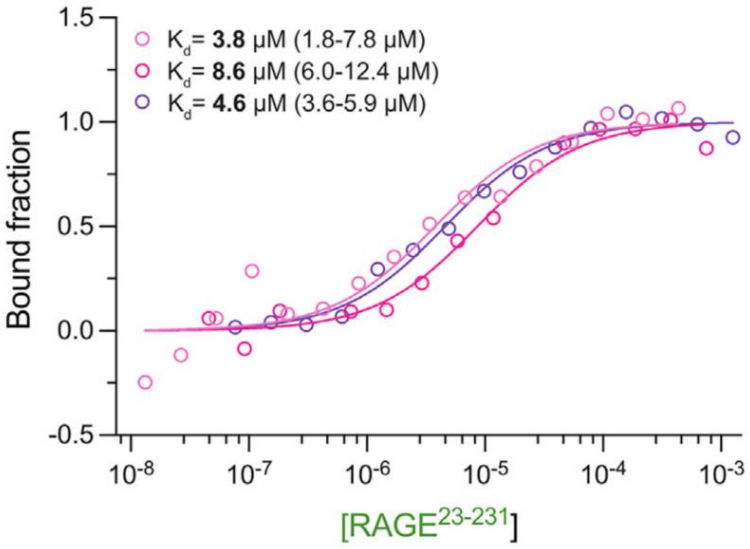
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The dissociation constant represents the concentration at which half of the proteins are occupied by the ligand:

When $[P] = [PL] \implies K_D = [L]$

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The binding affinity K_A is the reciprocal of the dissociation constant:

$$K_A = \frac{1}{K_D}$$

The dissociation constant is linked to the binding free energy ΔG (difference of free energy between the bound and unbound state) by the following equation:

$$\Delta G = RTlnK_D$$

$$K_D = e^{\Delta G/RT}$$

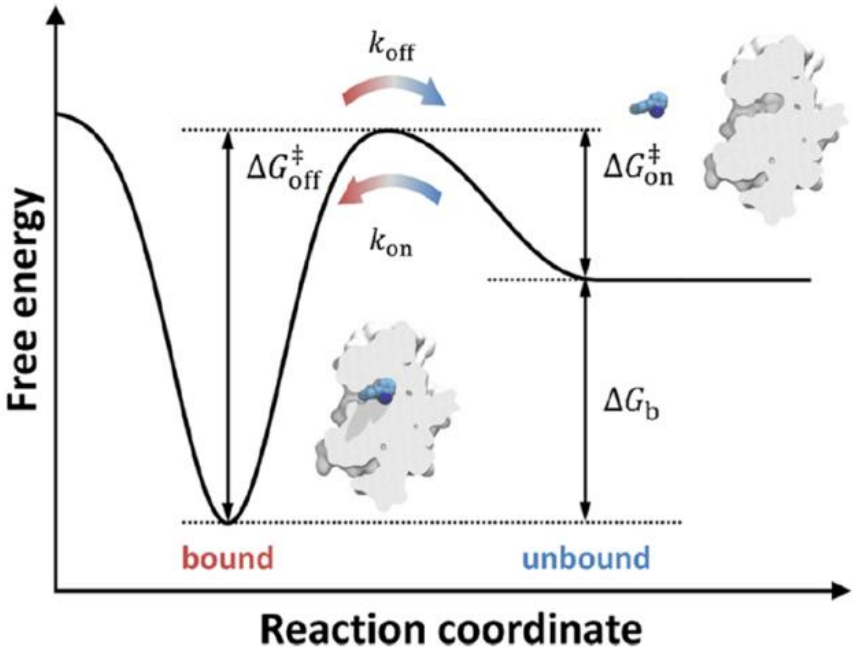
Energy measured in $KJ/mol \cdot K$

$$K_D = e^{\Delta G/K_B T}$$

Energy measured in $K_B T$

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Binding Affinity and Binding Free Energy



CHEMICAL REVIEWS
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Thermodynamics and Kinetics of Drug-Target Binding by Molecular Simulation

Sergio Decherchi and Andrea Cavalli*

Cite This: https://dx.doi.org/10.1021/acs.chemrev.0c00534 Read Online

$$K_D = e^{\Delta G/RT}$$

k_{on} and k_{off} are the rates at which one reaction and its reverse reaction happen.

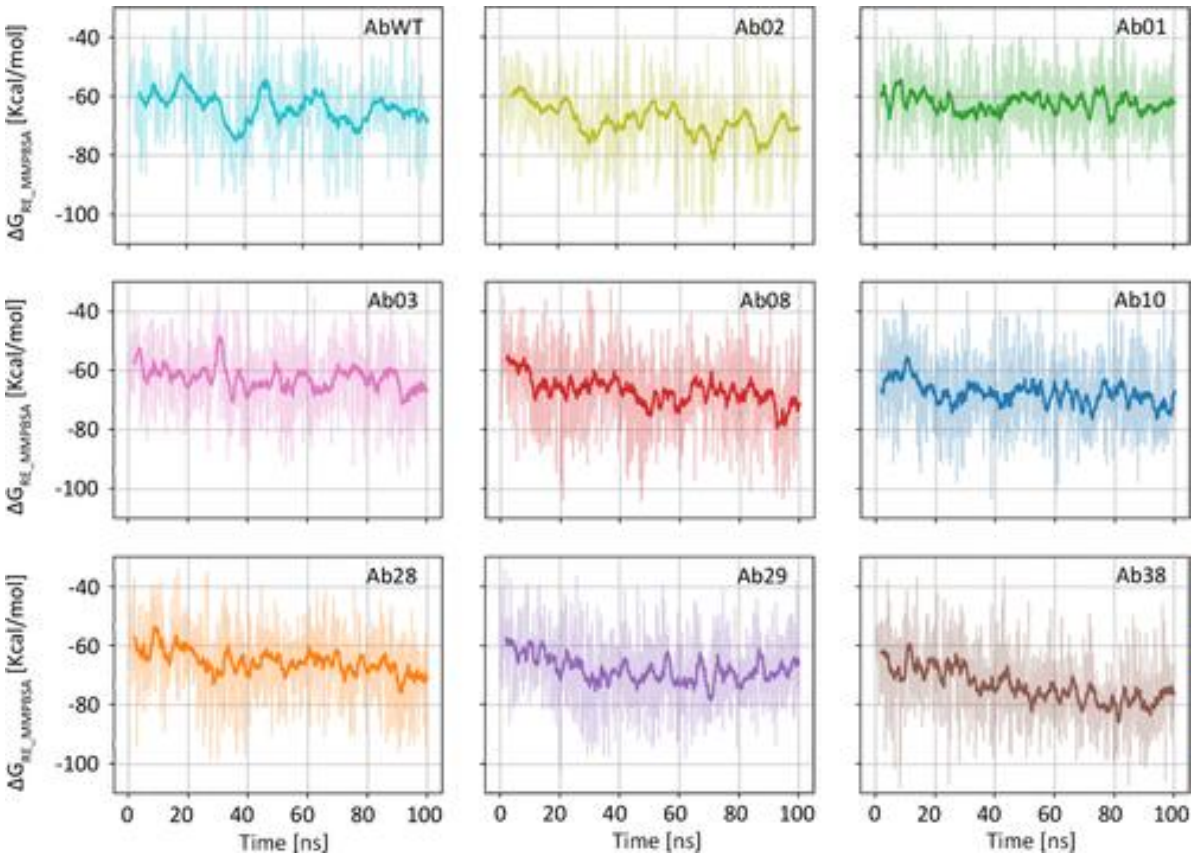
The rates are proportional to the probability of "jumping" the free energy barriers, which is the exponential of the free energies:

$$k_{off} \propto e^{\Delta G^{\ddagger}_{off}/RT} \; ; \; k_{on} \propto e^{\Delta G^{\ddagger}_{on}/RT}$$

$$K_D = \frac{k_{off}}{k_{on}} = e^{(\Delta G^{\ddagger}_{off} - \Delta G^{\ddagger}_{on})/RT} = e^{\Delta G/RT}$$

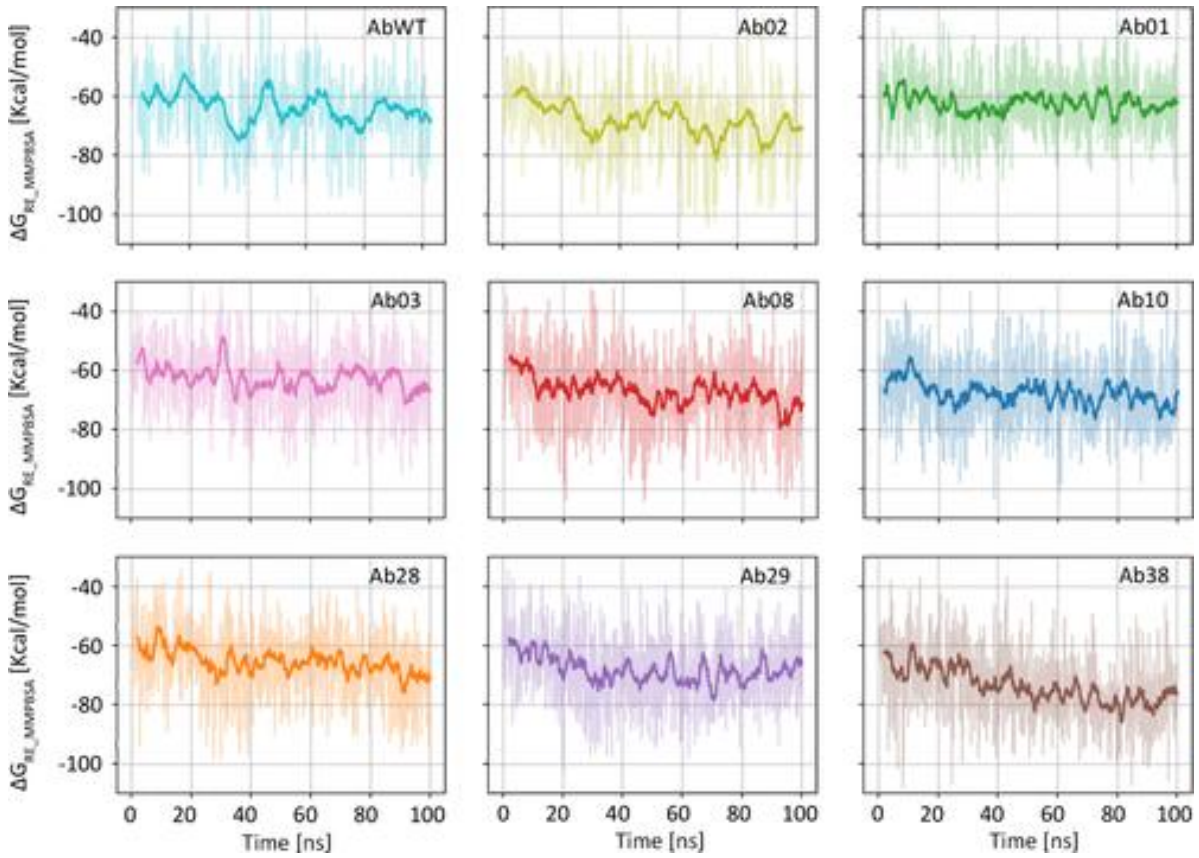
ENERGY MINIMIZATION

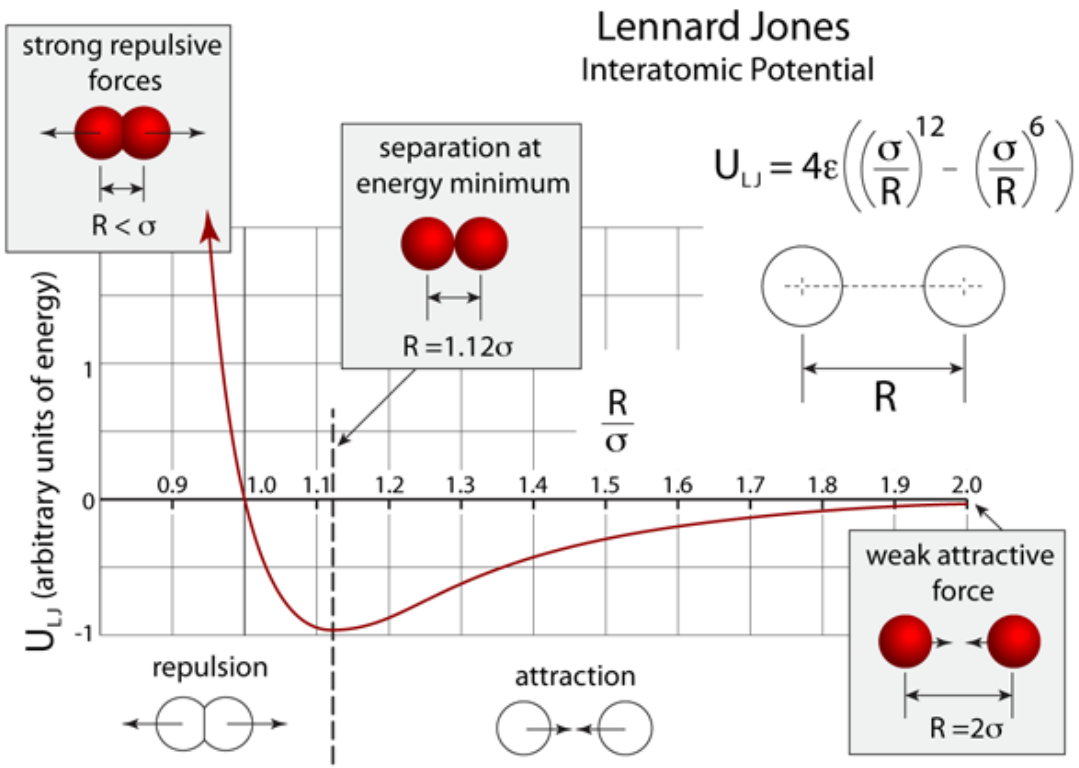
Why do we need Energy Minimization



Each graphs represent the binding energy of an antibody to a target, calculated using atomistic force fields

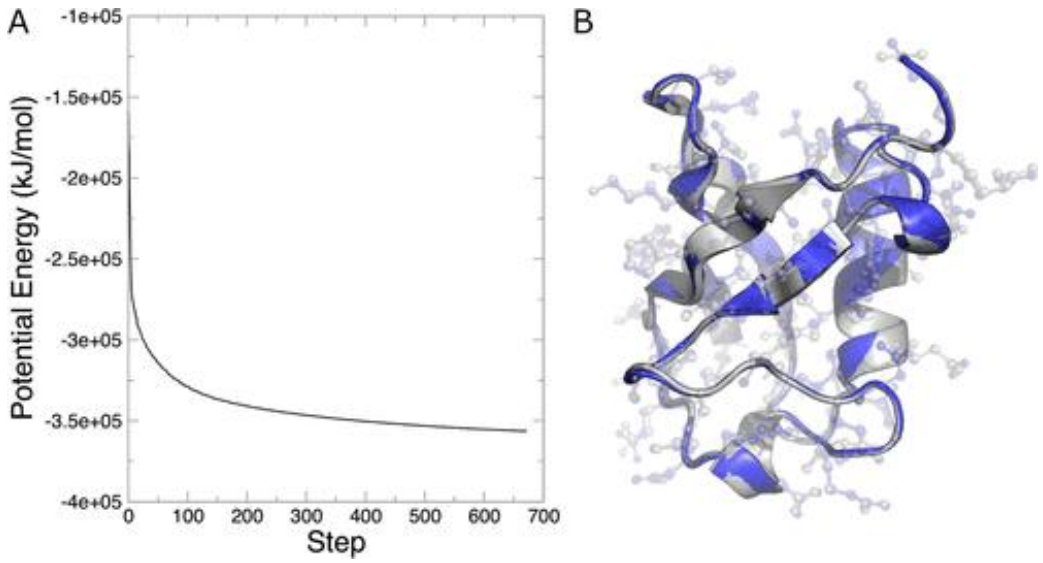
Why do we need Energy Minimization





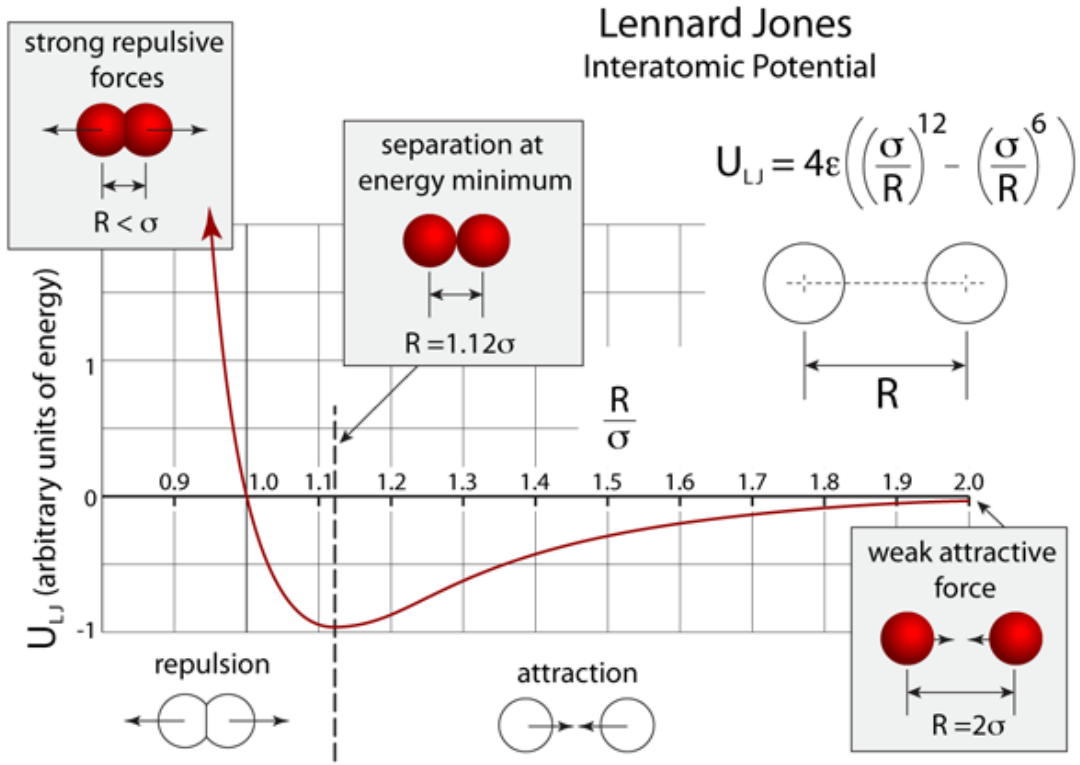
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Why do we need Energy Minimization

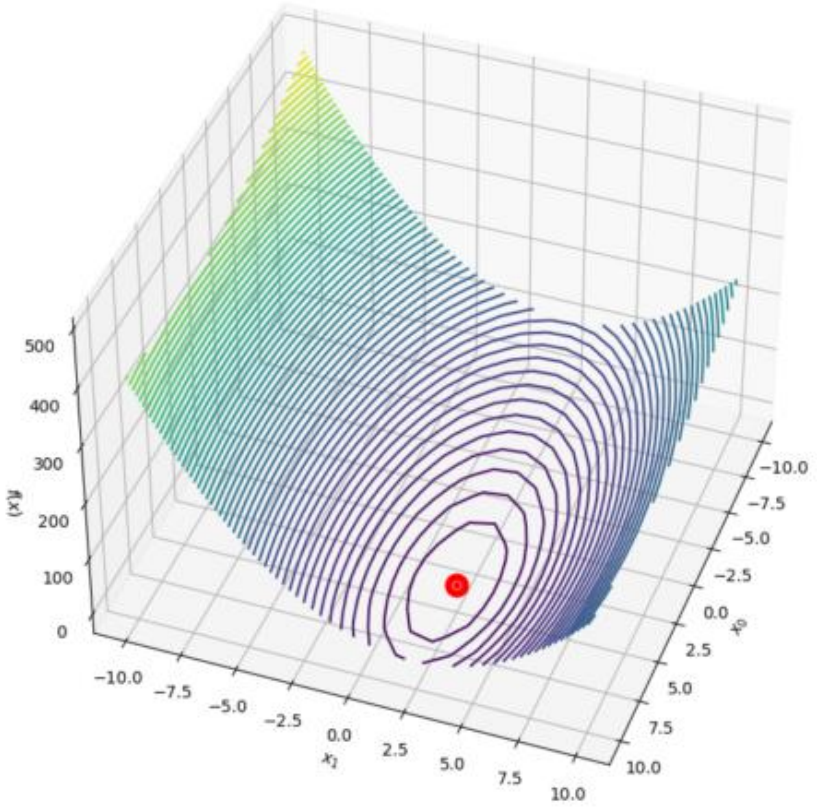


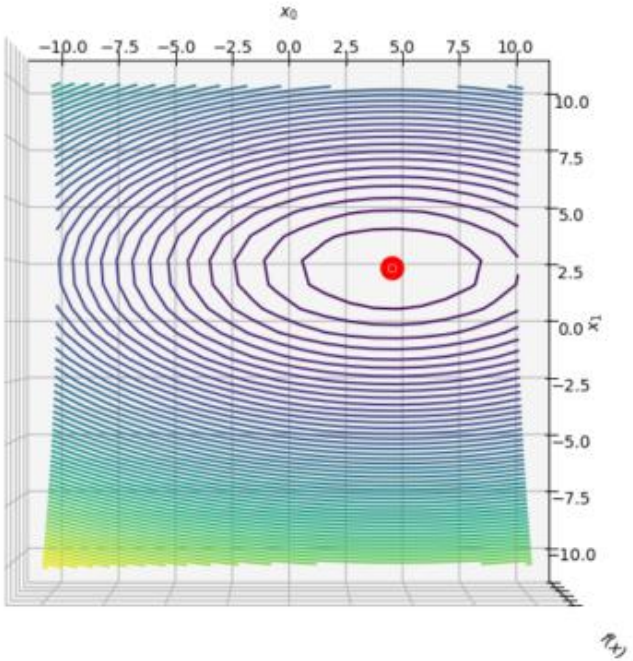
(A) Potential energy as a function of energy minimization step. (B) Overlay of the crystal structure of ubiquitin (gray) and the energy-minimized structure (blue).

Lemkul, The Journal of Physical Chemistry B Vol 128 Issue 39



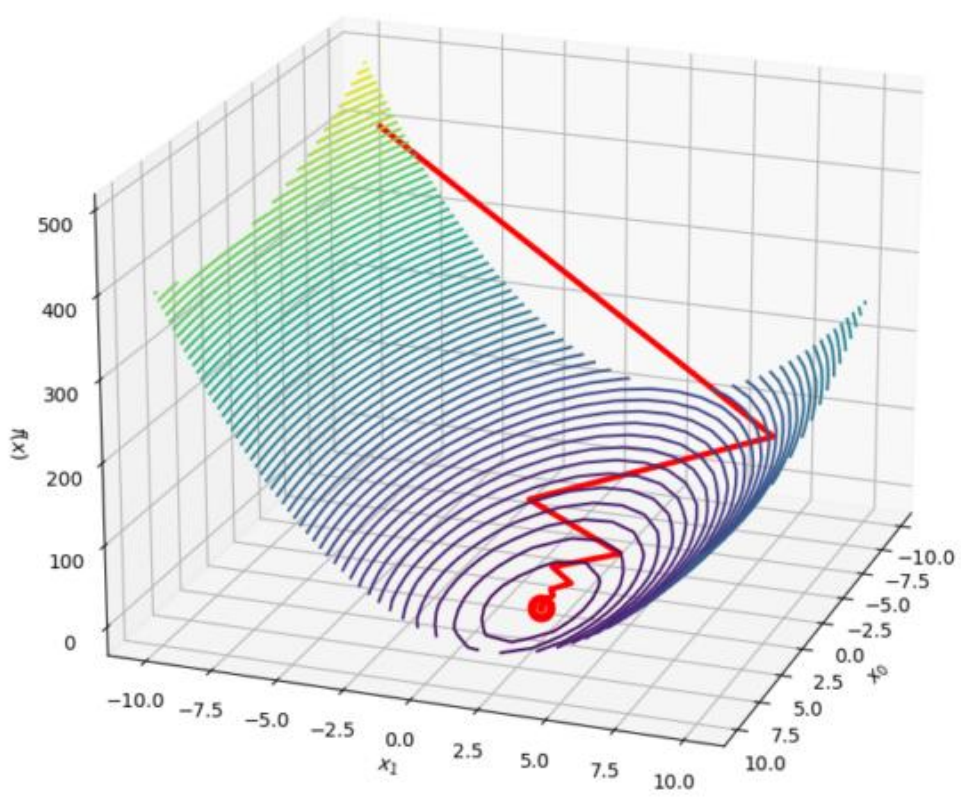
Example: Steepest descent

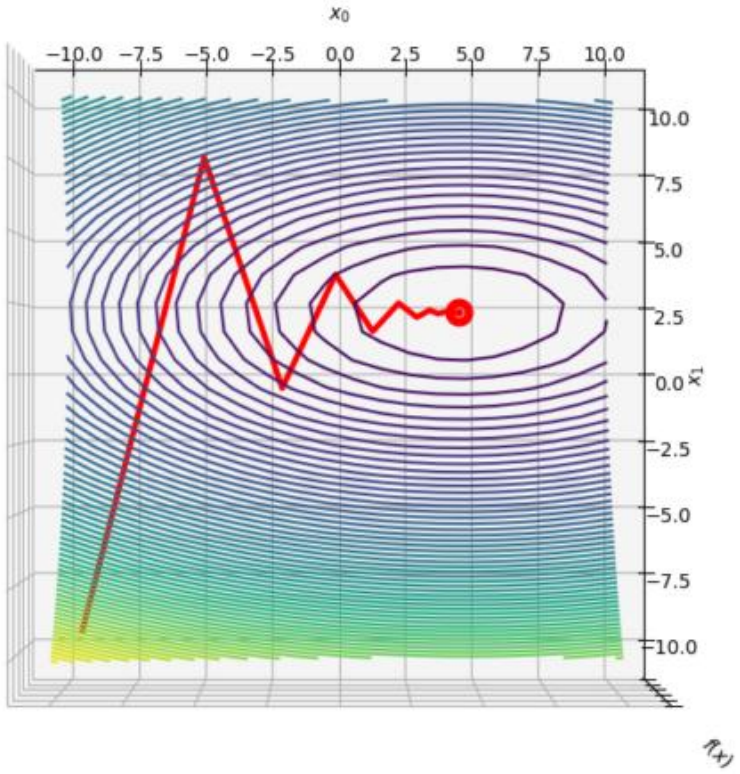




As an example, we want to find the minimum of this energy surface
(lines represent space regions having the same energy)

Example: Steepest descent





Starting from an arbitrary position $\boldsymbol{x_0}$ we define a successions of points:

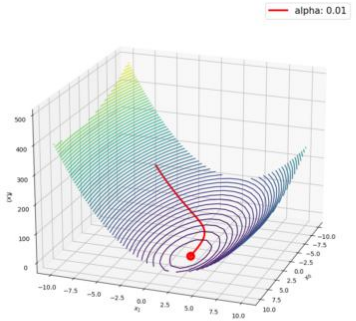
$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + a_k \boldsymbol{p}_k$$

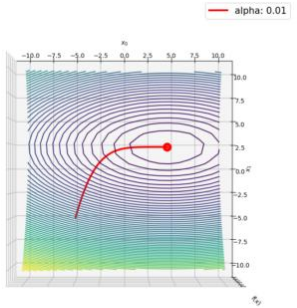
f is our energy function
$\boldsymbol{x}_k$ is the variable
$a_k > 0$ is the step size
$\boldsymbol{p}_k$ is the search direction

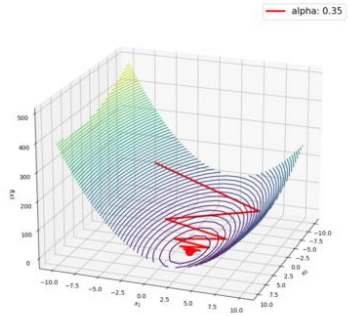
A sensitive choice for $\boldsymbol{p}_k$ is the opposite of the gradient of f:

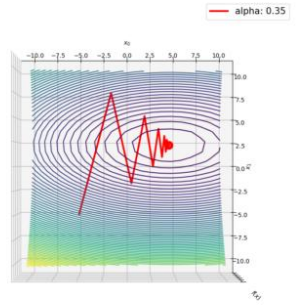
$$\boldsymbol{p}_k = -\nabla f(\boldsymbol{x}_k)$$

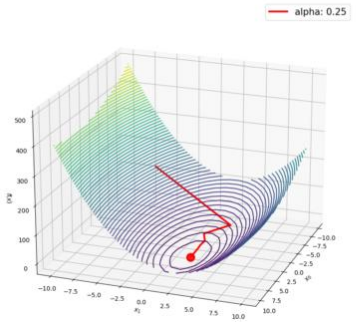
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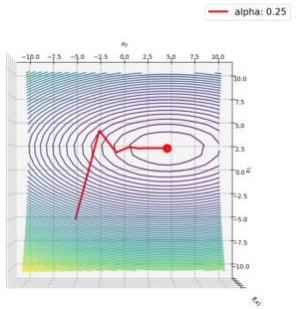


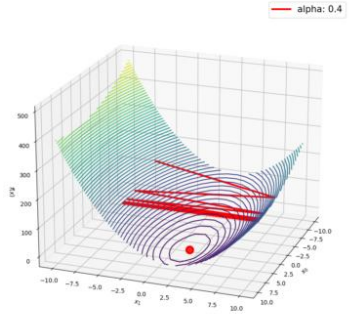


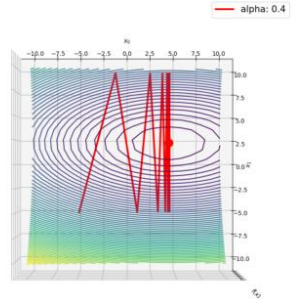


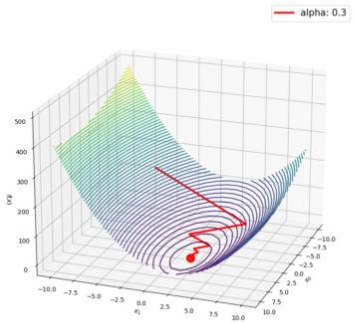


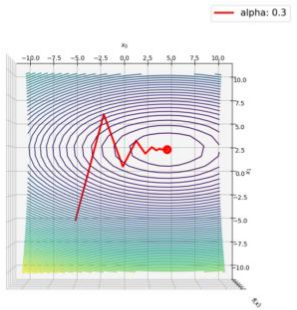












The step size needs to be carefully chosen to guarantee convergence.

It can be updated during the search.

Limitations of the steepest descent method

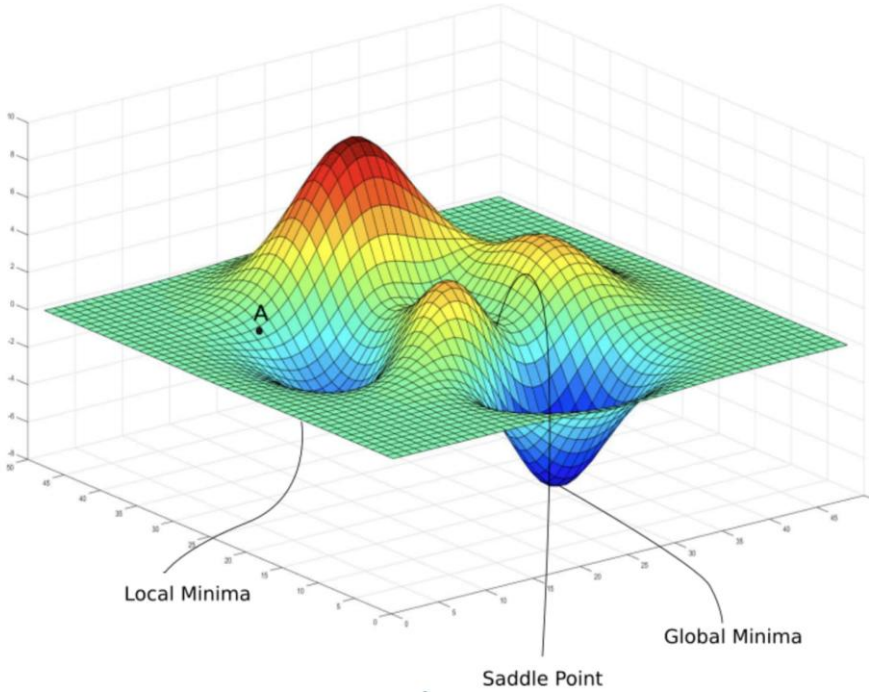
Local minima: The algorithm can converge to a local minimum instead of the global minimum, especially in non-convex optimization problems

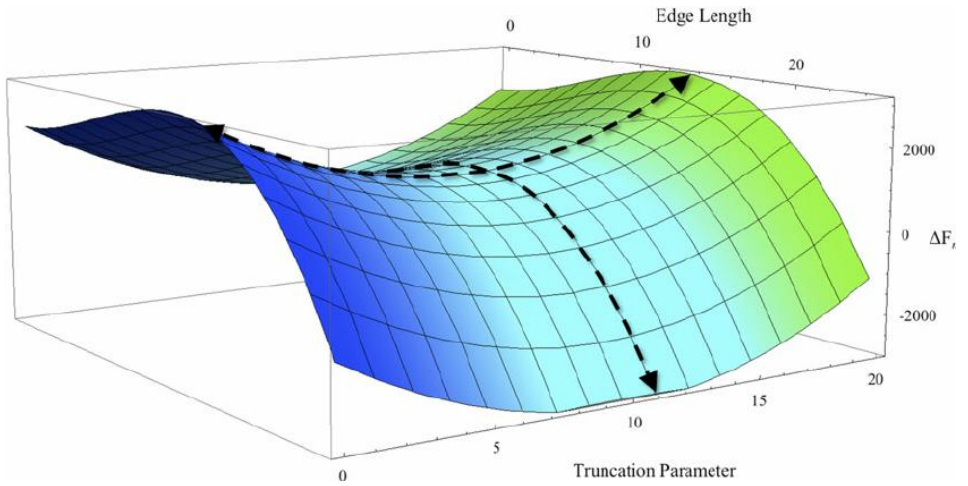
Saddle points: Can also get stuck at saddle points where the gradient is zero but it's neither a minimum nor maximum

Slow convergence: Can be very slow, especially near the minimum where gradients become small (the "ravine problem")

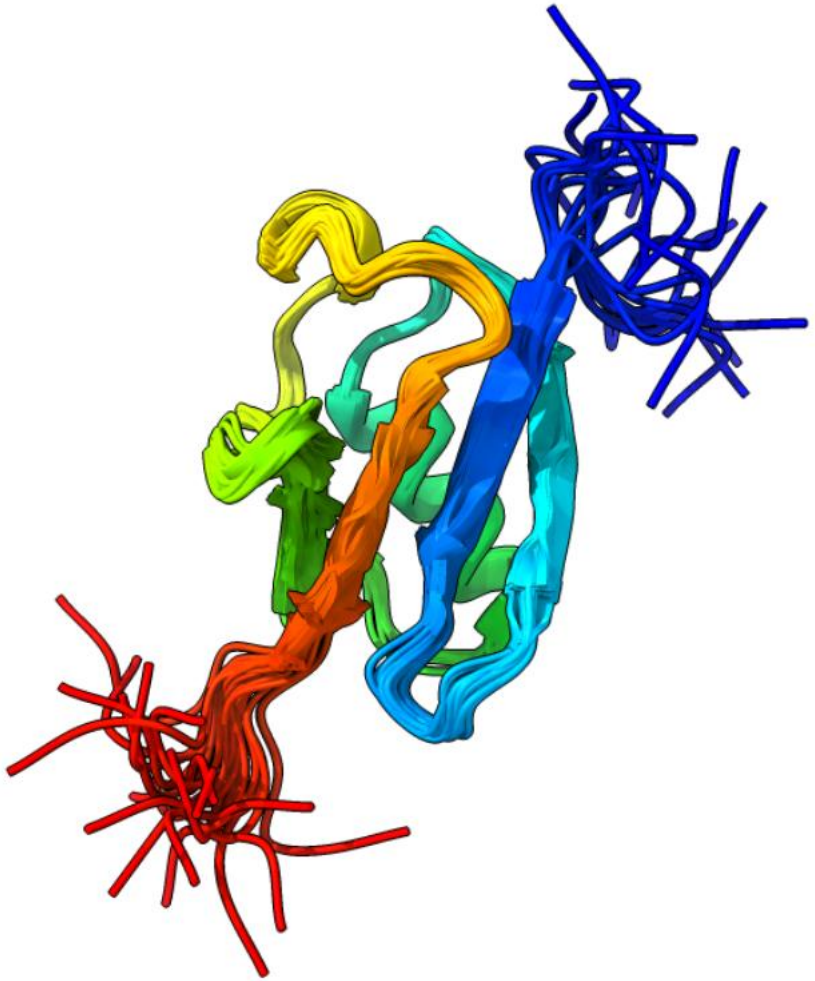
Learning rate sensitivity: Performance heavily depends on the choice of step size/learning rate — too small is slow, too large can overshoot or diverge

Plateau regions: Can make very slow progress in flat regions where gradients are near zero

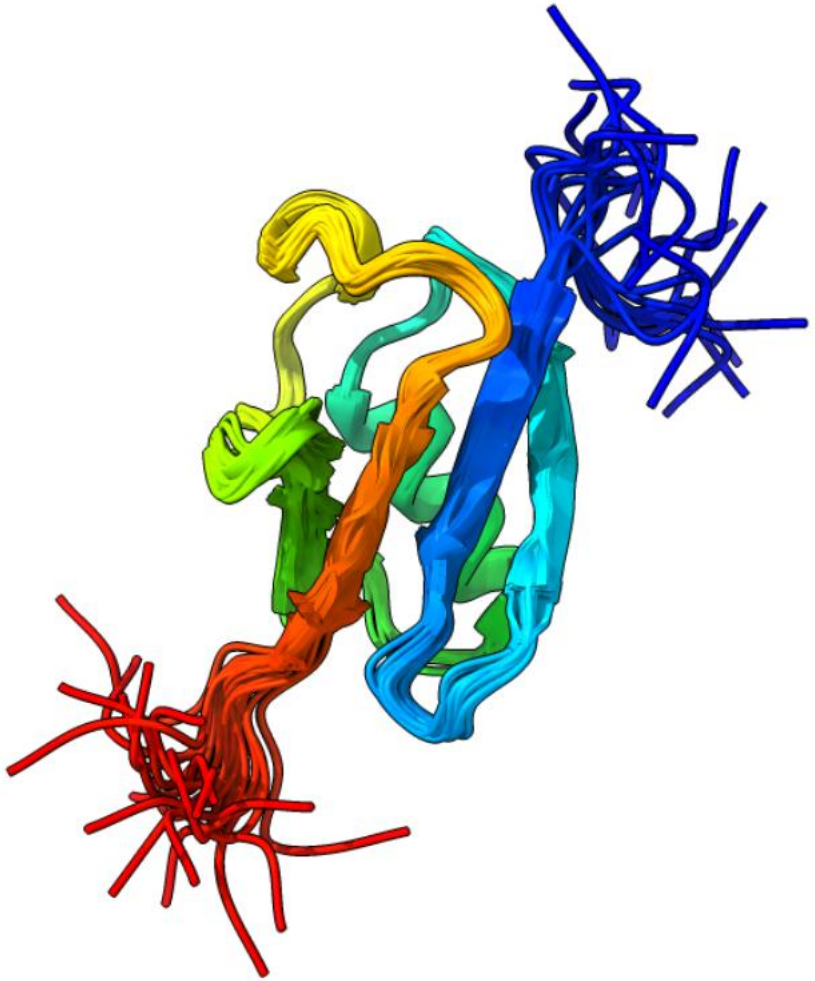




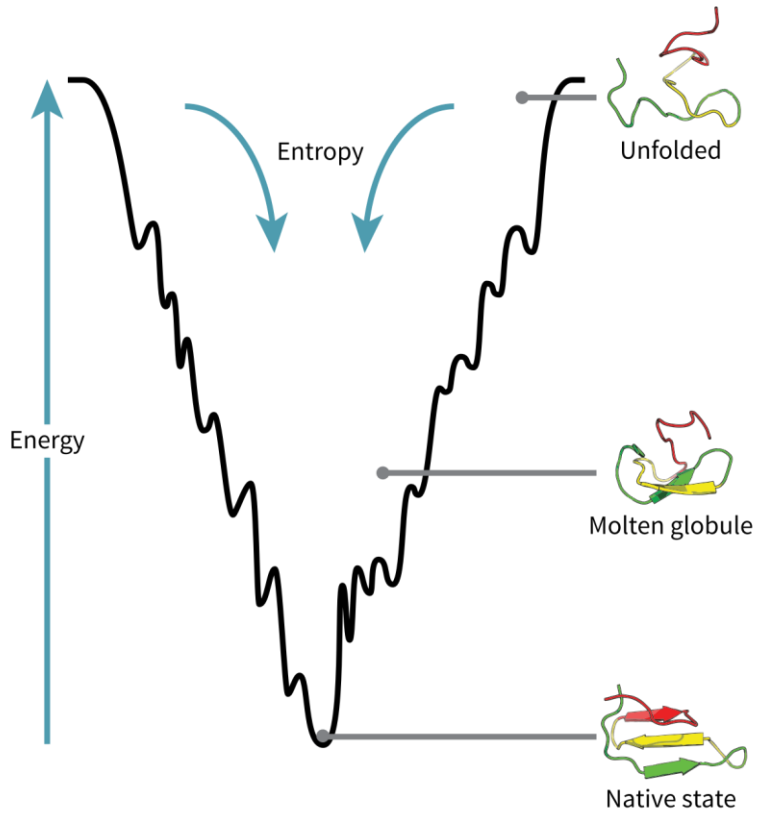
Is the minimum of energy the “best” configuration?



Is the minimum of energy the “best” configuration?



NMR ensemble for a small protein



The “energy funnel” can be very deep and narrow.

Nevertheless, **macromolecules are always subjected to thermal fluctuations**.

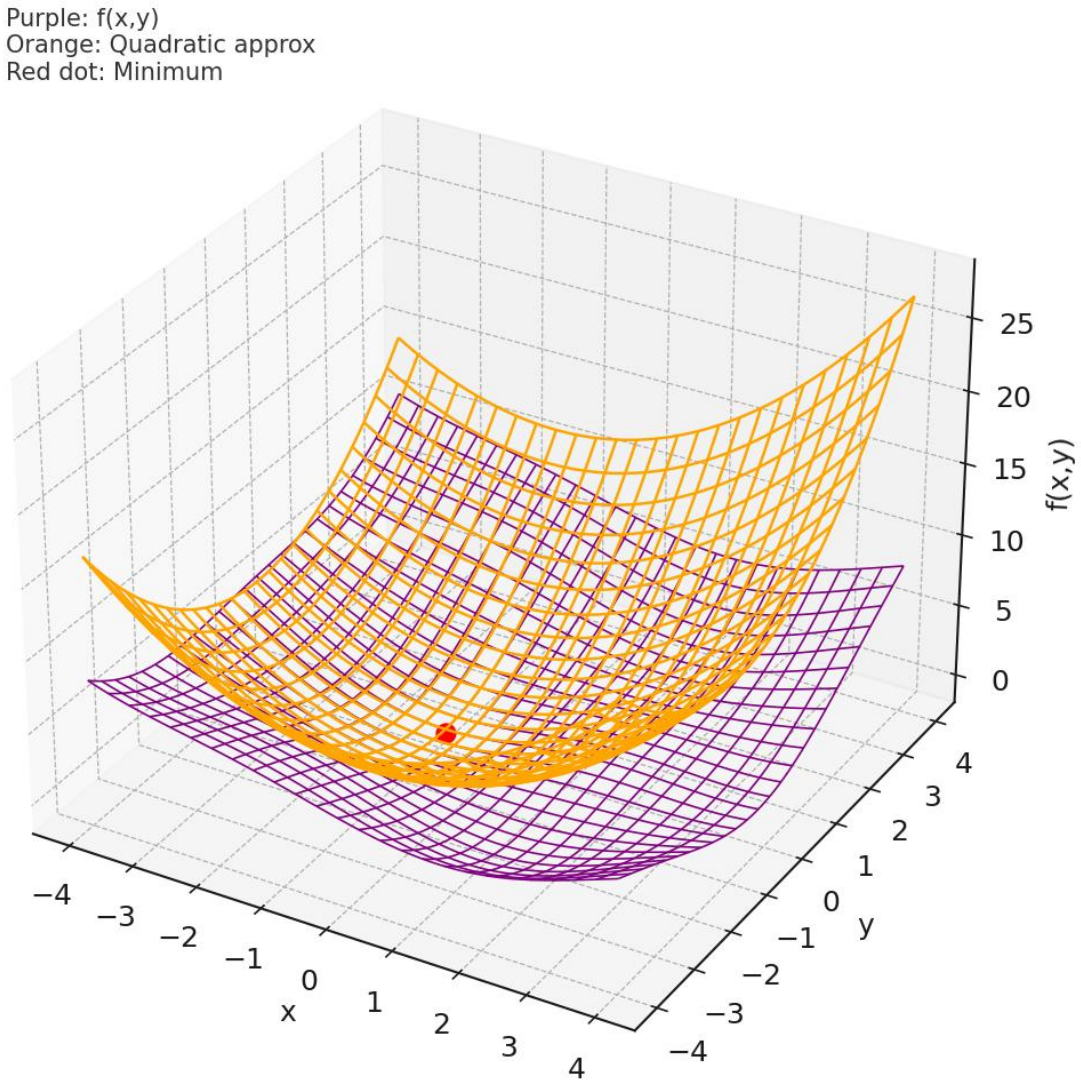
The minimum of energy represent a single representation of the “native” state.

Even if it is the most probable states**, many other states can be reached with the available thermal energy**.

MD simulations can be used to study a molecular system around the energy minimum (at equilibrium).

NORMAL MODES

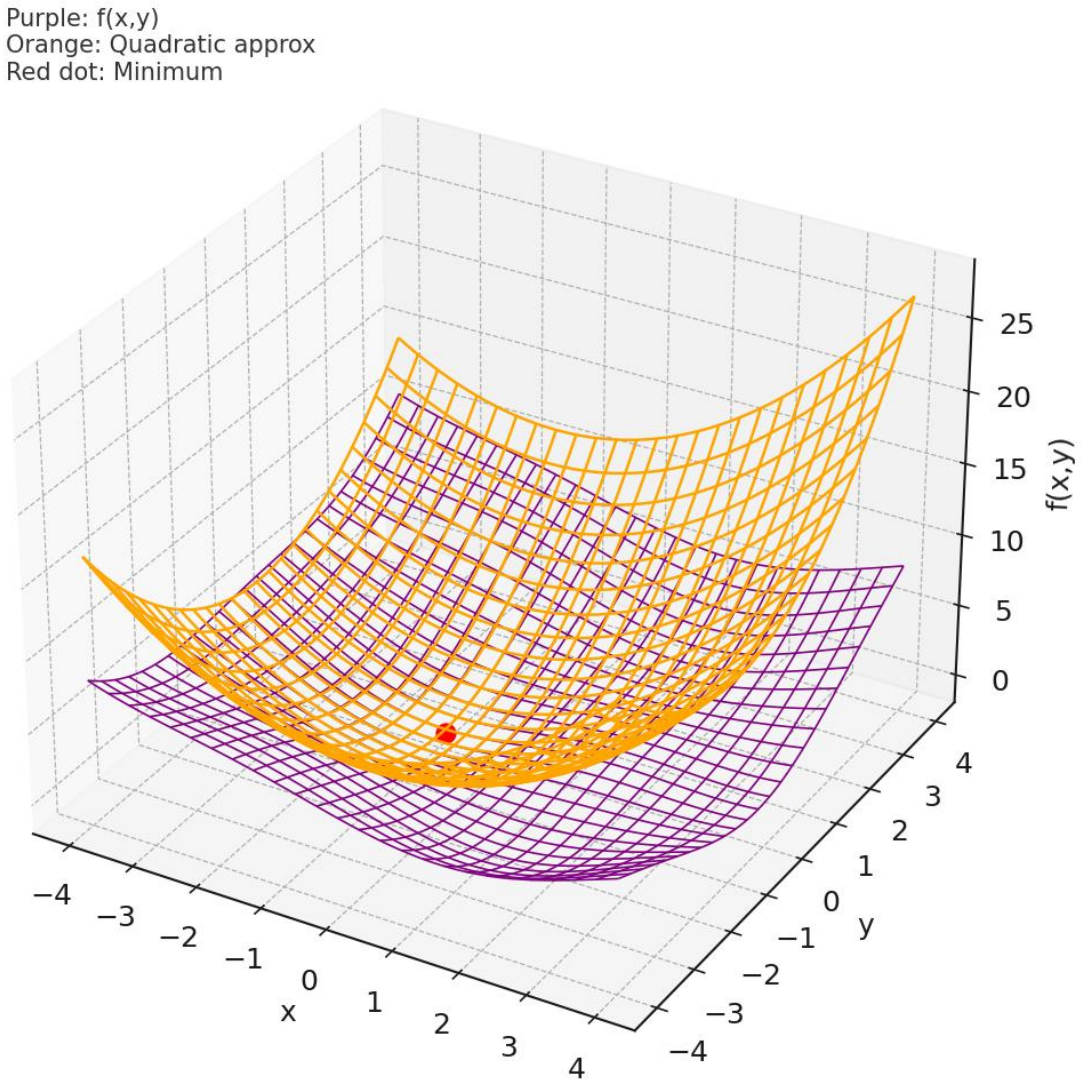
Approximation of energy near the minimum



Every function, even the most complicated, can be approximated near the minimum with a quadratic function (*).

(*) if the derivative exists!!!

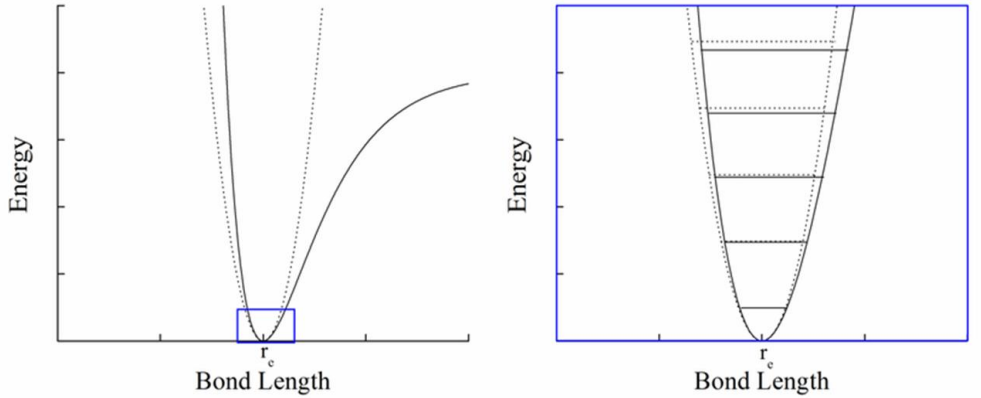
Approximation of energy near the minimum

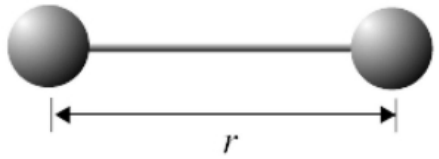


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(*) if the derivative exists!!!

This is why we could approximate chemical bonds and angles with harmonic oscillators.





$$U_r = k_r(r - r_0)^2$$

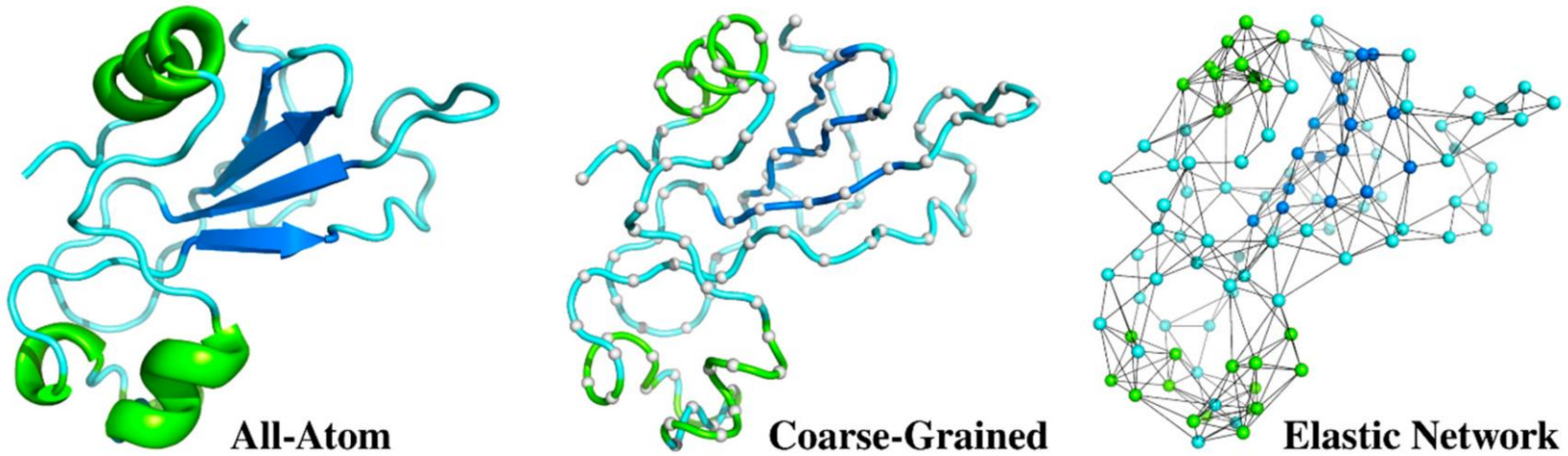
Elastic Network Models and Normal Modes Analysis

Introduced by Tirion, M. M. (1996), *Large amplitude elastic motions in proteins from a single-parameter, atomic analysis, Physical Review Letters* **77**, 1905–1908.

Elastic network models are highly simplified coarse-grained model in which (C-alpha) atoms are connected by an elastic potential if they are within a certain cut-off value (about 8 Å).

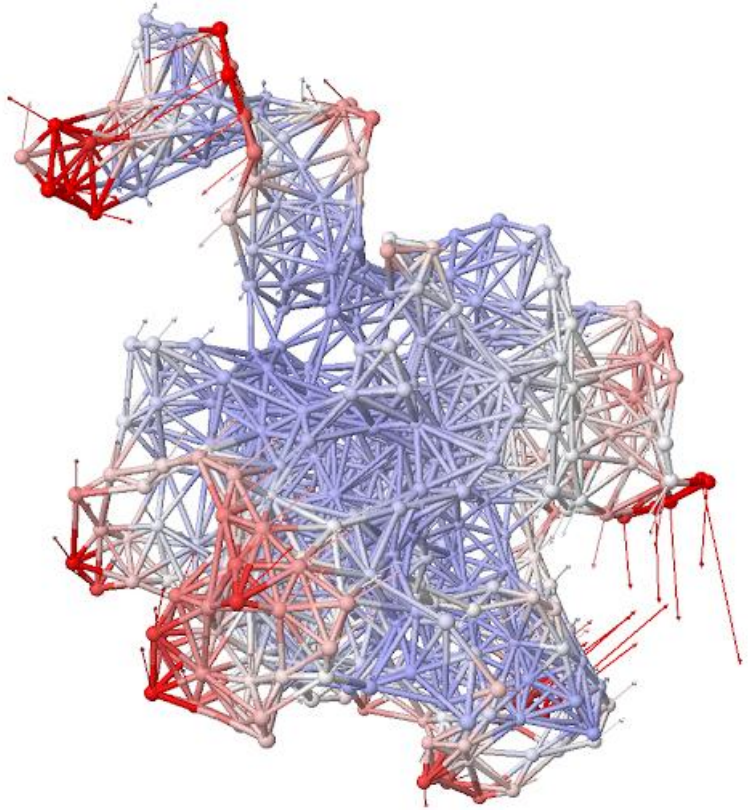
The dynamics of this model can be solved analytically producing a series of “Normal Modes”.
The general solution is a linear over position of Normal Modes.

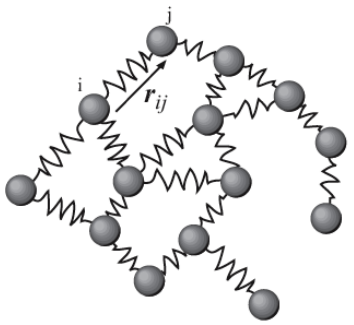
The Normal Modes are very similar to the Principal Components of the MD trajectories.



(a) (b) (c) Kmiemik et al. 2018

Anisotropic Network model and Hessian matrix





Simplified model: residues that are within a cut-off distance are connected with a spring.

$$f : R^n \rightarrow R$$

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & & & \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

From the energy we build the **Hessian matrix**. f represent the total energy in this model;

n is the total number of variables (n=3N, N number of residues).

Francesco Zonta – Room SB465, Department of Biosciences and Bioinformatics – francesco.zonta@xjtlu.edu.cn

Anisotropic Network model and Hessian matrix

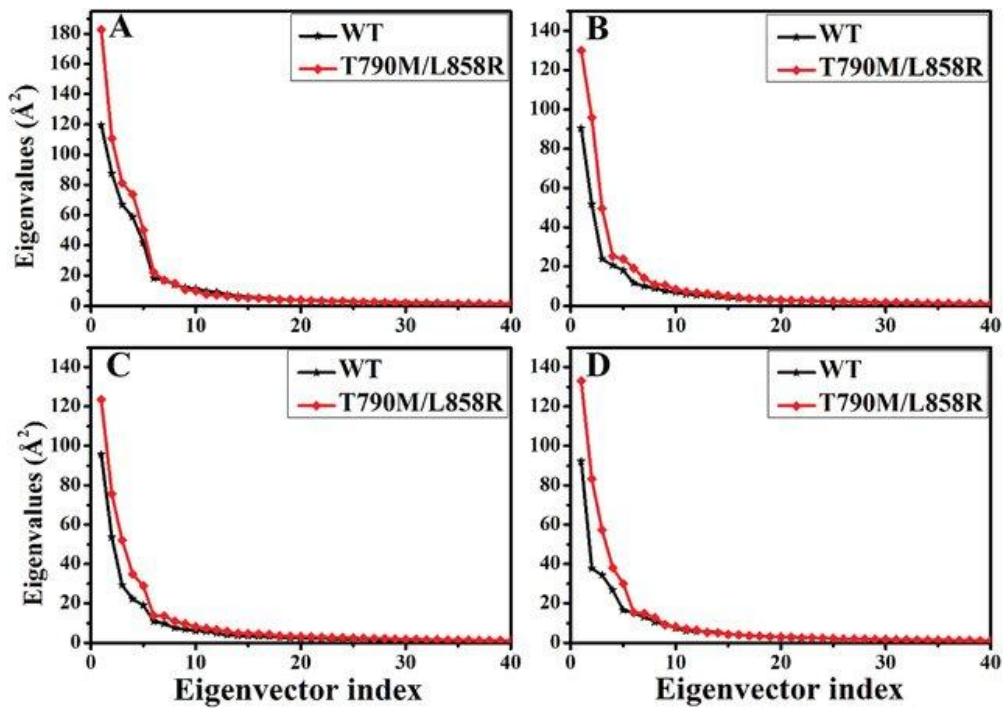
$$f : R^n \rightarrow R$$

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & & & \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

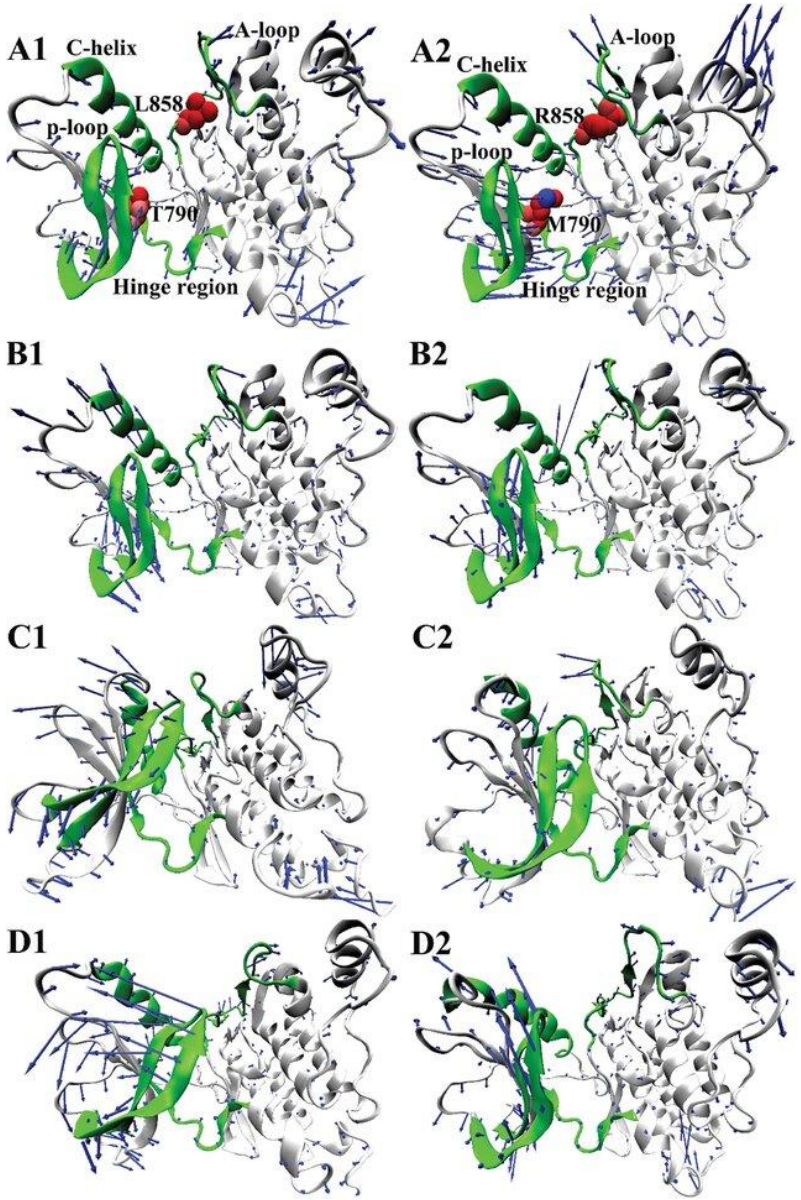
Diagonalization of the Hessian matrix.

Eigenvectors: coordinated motions of the residues

Eigenvalues: relative amplitude of the eigenvectors. The first eigenvalues dominates the overall motions.

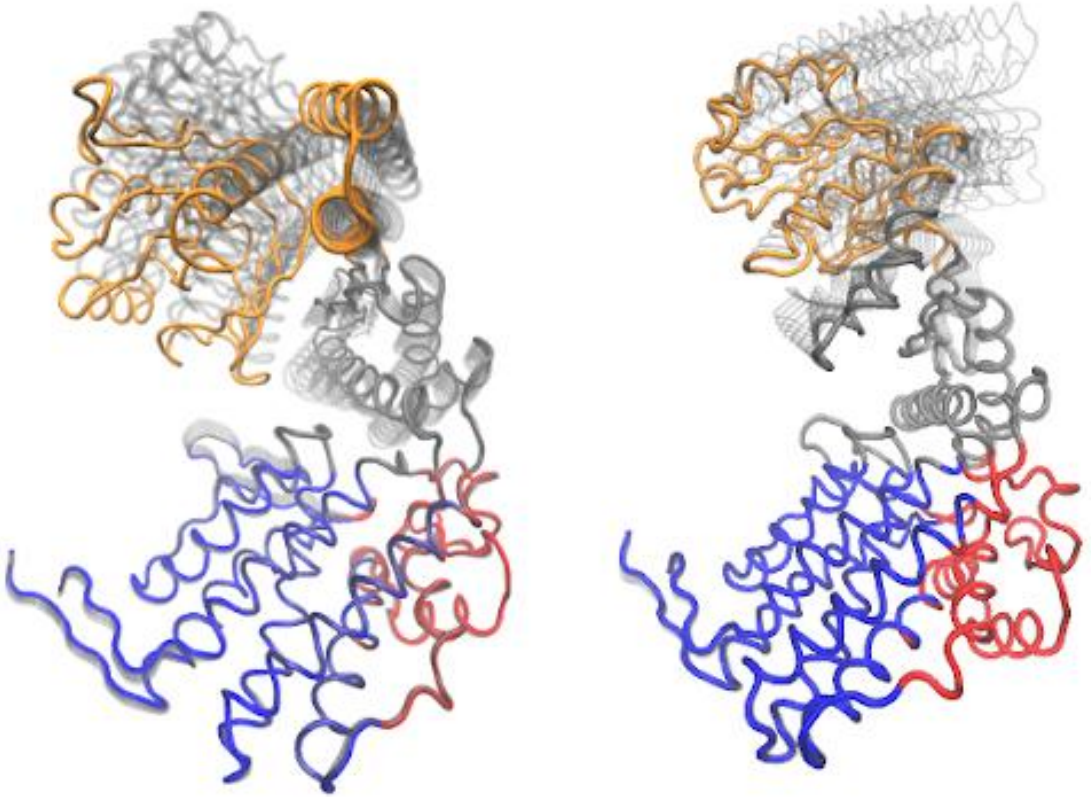


What to expect from elastic models



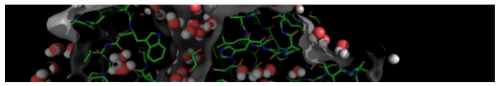
Fluctuations and eigenvectors observed from normal modes analysis correlate very well with the principal component analysis of MD trajectories.

This analysis can produce a reliable depiction of a protein at equilibrium at a much-reduced computational cost.



Webservers to try

Nantes Université



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Should you encounter any unexpected behaviour, please let us know.

Welcome to *elNémo* !

elNémo is the Web-interface to the *Elastic Network Model* (ENM), a fast and simple way for computing the low frequency normal modes of a macromolecule (Tirion, 1996; Sanejouand, 2012). Note that, thanks to the RTB approximation (Durand et al., 1994; Tama et al., 2000), this server can perform calculations for large all-atom systems.

[START A NEW RUN]

One major application of normal modes is the identification of potential conformational changes (Tama and Sanejouand, 2001; Delarue and Sanejouand, 2002; Mahajan and Sanejouand, 2015), e.g. of enzymes upon ligand binding. It has for instance been used in the study of membrane channel opening (Valadie et al., 2003) and for the analysis of structural movements of systems as large as the ribosome (Tama et al., 2003).

It has been shown that in 50% of the cases where protein structures are available in two different conformations, the related movement can be described by using only one or two low frequency normal modes (Krebs et al., 2002). This prompts for an application in X-ray cristallography data phasing, that is to use normal mode perturbed models as templates in molecular replacement (Suhre and Sanejouand, 2004).

Some applications of normal mode analysis are presented in the example section. In the reference section you may find a number of papers that describe the concepts behind this approach. To access previous jobs, go to the job status page.

The present version of ***elNémo*** allows you to compute the low frequency normal modes for a given protein structure in PDB format. You will be able to analyse these modes at different levels of detail, i.e. compare the collectivity of the modes, view 3-D animations of the protein movement for each mode and identify those residues that have the largest distance fluctuations in a given mode. Comparision of B-factors derived from the normal mode decomposition and measured B-factors gives an indication on differences in protein flexibility of the free protein and the protein in a crystallographic environment. When you submit two conformations of the same protein, you may analyse the contribution of each mode to the



ProDy

Protein Dynamics & Sequence Analysis

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Search Manual and Tutorials Go

ProDy Project

ProDy is a free and open-source Python package for protein structural dynamics analysis. It is designed as a flexible and responsive API suitable for interactive usage and application development.

Structure analysis

ProDy has fast and flexible PDB and DCD file parsers, and powerful and customizable atom selections for contact identification, structure comparisons, and rapid implementation of new methods.

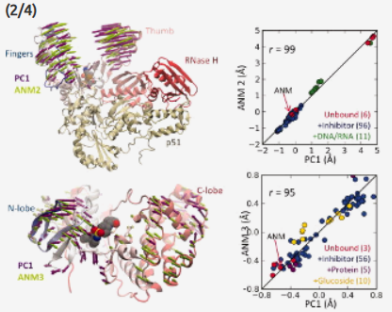
- New structure analysis tools include
  - InSty for analysing interactions and stability of proteins and protein-protein complexes
  - WatFinder for finding analysing water bridges

Dynamics analysis

- Principal component analysis can be performed for
  - heterogeneous X-ray structures (missing residues, mutations)
  - mixed structural datasets from Blast search
  - NMR models and MD snapshots (essential dynamics analysis)
- Normal mode analysis can be performed using
  - Anisotropic network model (ANM)
  - Gaussian network model (GNM)
  - ANM/GNM with distance and property dependent force constants

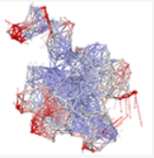
Dynamics from experimental datasets, theoretical models and simulations can be visualized using *NMWiz*.

Compare Dynamics from Experiments and Theory (2/4)

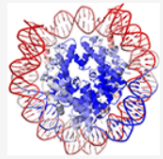


Comparative analysis of dynamics of drug target proteins and model systems from experiments (PCA) and theory (ANM). See the Protein Science article for details.

new ANM server



new iGNM database



References

Zhang S, Krieger JM, Zhang Y, Kaya C, Kaynak B, Mikulska-Ruminska K, Doruker P, Li H, Bahar I ProDy 2.0: Increased scale and scope after 10 years of protein dynamics modelling with Python **2021** *Bioinformatics* 37(20):3657-3659

Bakan A, Meireles LM, Bahar I ProDy: Protein Dynamics Inferred from Theory and Experiments **2011** *Bioinformatics* 27(11):1575-1577

Bakan A, Dutta A, Mao W, Liu Y, Chennubhotla C, Lezon TR, Bahar I Evol and ProDy for Bridging Protein Sequence Evolution and Structural Dynamics **2014** *Bioinformatics* 30(18):2681-2683